

Inferential Statistics: Knowledge Score vs. Residential Province

One-Way ANOVA (Analysis of Variance)

This report examines whether clinical knowledge differs significantly based on geographical residence, specifically comparing the Western Province, North Western Province, and all other provinces grouped together.

Descriptive Statistics

Province	N	Mean (V3 Score)	Std Dev (V3 Score)	Mean (Weighted V3)	Std Dev (Weighted)
Western	120	11.325	8.171	1.872	4.524
North Western	41	9.317	7.983	0.935	4.402
Other	40	13.425	7.900	3.337	4.636

Hypothesis Testing (One-Way ANOVA)

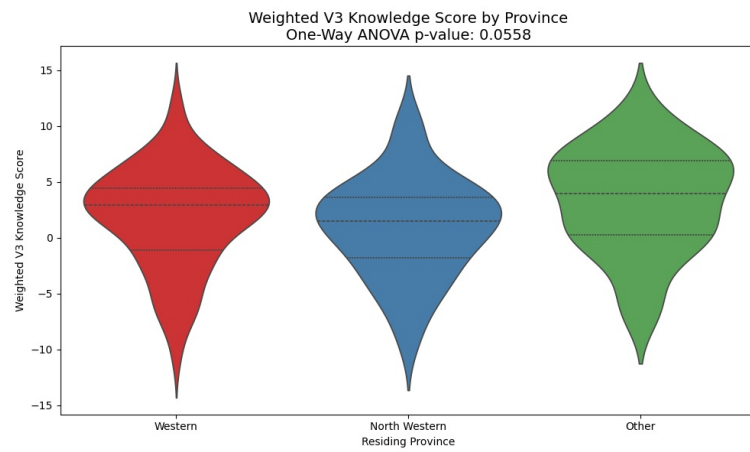
1. Analysis using the mathematically rigorous Weighted V3 Knowledge Score:

- **F-Statistic:** 2.929
- **P-Value:** 0.0558
- **Conclusion:** Not Statistically Significant ($p > 0.05$). Geography does not significantly impact knowledge.

2. Analysis using the standard V3 Knowledge Score:

- **F-Statistic:** 2.617
- **P-Value:** 0.0756
- **Conclusion:** Not Statistically Significant ($p > 0.05$).

Visualizations



Violin Plot